

In each of these cases, if only one CPU is listed, then you are not running SMP. Multiple CPUs in the listings indicate SMP mode.

Disabling SMP

In some situations, such as application benchmarking or hardware debugging, you may want to disable SMP support. This can be done with the kernel parameters `nosmp` or `maxcpus=1`. If this is a temporary need, you can just boot the system, catch Grub at the menu by pressing ESC, and typing `boot nosmp maxcpus=1` at the prompt. If you have multiple boot options, then you may need to edit the kernel line and add `nosmp maxcpus=1` to the kernel boot line.

For a longer-term solution, consider adding these boot parameters to the Grub configuration.

1. As root, edit `/boot/grub/menu.lst`.

2. Scroll down to your kernel.

3. Add a kernel option for disabling SMP. For example:

```
kernel /boot/vmlinuz-2.6.15-26-686 root=/dev/hda1 ro  
splash maxcpus=1
```

5. Save your changes.

The next reboot will use your new changes.

Note: If you modify `/boot/grub/menu.lst`, be aware that the contents could be overwritten the next time you upgrade the kernel or run `update-grub`.

Missing SMP?

If you find that you only have one active CPU on a multiple CPU system, try installing the explicit SMP kernel: `sudo apt-get install linux-686-smp`. Beyond that, there are few generic debugging options and the problem is unlikely related to Ubuntu—it is probably a general Linux kernel problem.

Check with the motherboard manufacturer and see if Linux supports their chipset. For example, we have an old dual-CPU motherboard that is not supported by Linux.

Check the Linux Hardware FAQ for the motherboard or chipset. This will tell you if other people managed to get it to work. Web sites such as <https://wiki.ubuntu.com/HardwareSupport> and <http://www.faqs.org/docs/Linux-HOWTO/SMP-HOWTO.html> are good places to start.

If all else fails, post a query to any of the Linux or Ubuntu hardware forums. Maybe someone else knows a workaround. Some good forums include <http://www.ubuntuforums.org/>, <http://www.linuxhardware.org/>, and <http://www.linuxforums.org/forum/>. Be sure to include details such as the make and model of the motherboard, Ubuntu version, and other peripherals. It is